

Math 428: Homework 6 Solutions

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Section 5.2: 4

Solution.

$$\int_0^\infty \frac{x^2}{(1+x^2)^2} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{x^2}{(1+x^2)^2} dx = \pi i \operatorname{Res}\left(\frac{z^2}{(1+z^2)^2}, i\right)$$

$$\operatorname{Res}\left(\frac{z^2}{(z-i)^2(z+i)^2}, i\right) = \left(\frac{z^2}{(z+i)^2}\right)' \Big|_{z=i} = \frac{2i}{(2i)^2} - \frac{2 \cdot i^2}{(2i)^3} = \frac{1}{4i}$$

$$\int_0^\infty \frac{x^2}{(1+x^2)^2} dx = \frac{\pi}{4}$$

Section 5.2: 5

Solution. $z^2+2z+2 = (z+1)^2+1$, roots are $z_1 = -1+i$, $z_2 = -1-i$.

$$\int_{-\infty}^\infty \frac{x}{(x^2+2x+2)^2} dx = 2\pi i \operatorname{Res}\left(\frac{z}{(z^2+2z+2)^2}, z_1\right)$$

$$\begin{aligned} \operatorname{Res}\left(\frac{z}{(z-z_1)^2(z-z_2)^2}, z_1\right) &= \left(\frac{z}{(z-z_2)^2}\right)' \Big|_{z=z_1} \\ &= \frac{1}{(z_1-z_2)^2} - \frac{2z_1}{(z_1-z_2)^3} \\ &= \frac{1}{(2i)^2} - \frac{-2+2i}{(2i)^3} \\ &= \frac{2}{(2i)^3} = \frac{i}{4} \end{aligned}$$

$$\int_{-\infty}^\infty \frac{x}{(x^2+2x+2)^2} dx = -\frac{\pi}{2}$$

5.2 #6

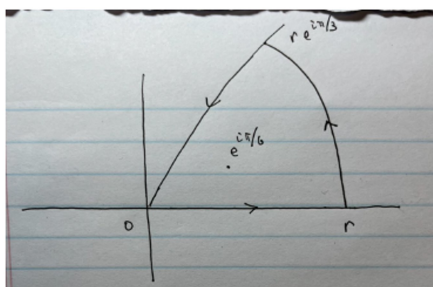
Proof. Let $\gamma_r = [0, x] + re^{it} - se^{i\pi/3}$ for $x \in [0, r]$, $t \in [0, \pi/3]$ and $s \in [0, r]$ as drawn in the picture. The line integral along γ_r is

$$\begin{aligned} \int_{\gamma_r} \frac{1}{1+z^6} dz &= \int_0^r \frac{1}{1+x^6} dx + \int_0^{\pi/3} \frac{1}{1+(re^{it})^6} rie^{it} dt - \int_0^r \frac{1}{1+(se^{i\pi/3})^6} e^{i\pi/3} ds \\ &= (1 - e^{i\pi/3}) \int_0^r \frac{1}{1+x^6} dx + \int_0^{\pi/3} \frac{1}{1+r^6 e^{i6t}} rie^{it} dt. \end{aligned}$$

By Residue theorem $\int_{\gamma_r} \frac{1}{1+z^6} dz = 2\pi i \text{Res}(f, e^{i\pi/6}) = -\frac{\pi i}{3} e^{i\pi/6}$. When r large enough, it is clear $|1 + r^6 e^{i6t}| \geq r^6 - 1 \geq r^6/2$. Then by ML-estimate, we have

$$\left| \int_0^{\pi/3} \frac{1}{1+r^6 e^{i6t}} rie^{it} dt \right| \leq \int_0^{\pi/3} \frac{1}{|1+r^6 e^{i6t}|} r dt \leq \frac{2}{r^5} \frac{\pi r}{3} \rightarrow 0 \text{ as } r \rightarrow \infty.$$

Taking $r \rightarrow \infty$, we have $-\frac{\pi i}{3} e^{i\pi/6} = \lim_{r \rightarrow \infty} \int_{\gamma_r} f(z) dz = (1 - e^{i\pi/3}) \int_0^\infty \frac{1}{1+x^6} dx$ □



Section 5.3: 2

Solution. I use the definition without the factor $1/\sqrt{2\pi}$.

If $s > 0$, then since $\text{Re} -isz = s \text{Im}(z)$ is negative if $\text{Im}(z) < 0$, we complete the contour in $\text{Im}(z) < 0$ (which will be clockwise hence index -1 at the pole) and

$$\int_{-\infty}^{\infty} \frac{z e^{-isz}}{1+z^2} dz = -2\pi i \text{Res}\left(\frac{z e^{-isz}}{1+z^2}, -i\right) = -\pi i e^{-s}, \quad s > 0.$$

If $s < 0$ we complete the contour in $\text{Im}(z) > 0$, and

$$\int_{-\infty}^{\infty} \frac{z e^{-isz}}{1+z^2} dz = 2\pi i \text{Res}\left(\frac{z e^{-isz}}{1+z^2}, i\right) = \pi i e^s, \quad s < 0.$$

Section 5.3: 5

Solution. $z^2 + 4z + 5 = (z+2)^2 + 1$ has simple roots $z_1 = -2 + i$ in the upper half space, and $z_2 = -2 - i$ in the lower half space. For $s \geq 0$ we proceed as above and write

$$\hat{f}(s) = \int_{[-\infty, \infty]} \frac{e^{-isz}}{z^2 + 4z + 5} dz = -2\pi i \text{Res}\left(\frac{e^{-isz}}{z^2 + 4z + 5}, z_2\right)$$

Since $z_2 - z_1 = -2i$,

$$\hat{f}(s) = -2\pi i \frac{e^{-s+2is}}{-2i} = \pi e^{-s} e^{2is}.$$

For $s \leq 0$, we can use $\hat{f}(s) = \overline{\hat{f}(-s)}$ to write, for all s ,

$$\hat{f}(s) = \pi e^{-|s|} e^{2is}.$$

5.3 #6

Solution.

$$\begin{aligned}
 \hat{f}(t) &= \int_{-\infty}^{\infty} e^{-itx} f(x) dx = \int_{-\infty}^{\infty} e^{-itz} \frac{\sin z}{z} dz = \int_{-\infty}^{\infty} e^{-itz} \frac{e^{-iz} - e^{iz}}{i2z} dz \\
 &= \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R)} e^{-itz} \frac{e^{iz} - e^{-iz}}{i2z} dz \\
 &= \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R)} \frac{e^{-i(t-1)z} - e^{-i(t+1)z}}{i2z} dz,
 \end{aligned}$$

where $\gamma_r = \{re^{i\theta} : \pi \leq \theta \leq 2\pi\}$ is the lower semi-circle of radius r running in the counter-clockwise direction.

Let $\Gamma_R^+ = \{Re^{i\theta} : 0 \leq \theta \leq \pi\}$ be the upper semi-circle of radius R running in the counter-clockwise direction. and $\Gamma_R^- = \{Re^{-i\theta} : 0 \leq \theta \leq \pi\}$ be the lower semi-circle of radius R running in the clockwise direction. Note that for $s \in \mathbb{R}$, $|e^{-isz}| = e^{s\text{Im}z}$. By the bounded convergence theorem, we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R^-} \frac{e^{-isz}}{z} dz = 0 \quad \text{if } s > 0, \quad (1)$$

and

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R^+} \frac{e^{-isz}}{z} dz = 0 \quad \text{if } s < 0, \quad (2)$$

So when $t > 1$, we have by (1) and Cauchy integral theorem,

$$\hat{f}(t) = \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R) \cup \Gamma_R^-} \frac{e^{-i(t-1)z} - e^{-i(t+1)z}}{i2z} dz = 0$$

When $t < -1$, we have by (2) and the residues theorem,

$$\begin{aligned}
 \hat{f}(t) &= \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R) \cup \Gamma_R^+} \frac{e^{-i(t-1)z} - e^{-i(t+1)z}}{i2z} dz \\
 &= 2\pi i \text{Res} \left(\frac{e^{-i(t-1)z} - e^{-i(t+1)z}}{i2z}, 0 \right) = 0.
 \end{aligned}$$

When $-1 < t < 1$, we have by (1)-(2), Cauchy integral theorem and the residues theorem,

$$\begin{aligned}
 \hat{f}(t) &= \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R) \cup \Gamma_R^+} \frac{e^{-i(t-1)z}}{i2z} dz \\
 &\quad - \lim_{R \rightarrow \infty, r \rightarrow 0} \int_{(-R, -r) \cup \gamma_r \cup (r, R) \cup \Gamma_R^-} \frac{e^{-i(t+1)z}}{i2z} dz \\
 &= 2\pi i \text{Res} \left(\frac{e^{-i(t-1)z}}{i2z}, 0 \right) - 0 \\
 &= \pi.
 \end{aligned}$$

5.3 #7

Proof. By definition $\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-itx}dx$. By integration by parts, we have

$$\hat{f}(t) = -\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{1}{it}e^{-itx}f'(x)dx = -\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} + \frac{1}{it}\hat{f}'.$$

Since $x, t \in \mathbb{R}$, by assumption $\lim_{x \rightarrow \infty} |e^{-itx}f(x)| = \lim_{x \rightarrow \infty} |e^{-itx}||f(x)| = \lim_{x \rightarrow \infty} |f(x)| = 0$, which implies $\lim_{x \rightarrow \infty} e^{-itx}f(x) = 0$. Similarly $\lim_{x \rightarrow -\infty} e^{-itx}f(x) = 0$. Thus $-\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} = 0$ and $\hat{f}(t) = \frac{1}{it}\hat{f}'$. $\square$